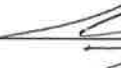


16. Sign-off Checklist

Checkpoint	Status / Remarks
Email / approved group-mail intake and document-classification rules confirmed	 <u>Vishey</u> K & Pader
Purchase-side extraction, portal entry and Loading Sheet synchronization rules validated	 <u>Vishey</u>
Sales-side invoice / contract generation including draft B/L handling and approval rules confirmed	 <u>Vishey</u> K & Pader
FA document types, data fields, validation and approval rules confirmed	
Container, B/L and shipment-tracking rules confirmed	 <u>Vishey</u> K & Pader
Mandatory documents and tolerance rules approved by business	 <u>Vishey</u>
Exception owners, duplicate / repeated-information handling, escalation and ageing rules approved	 <u>Vishey</u> K & Pader
Web portal as primary operational interface and tracker transition approach confirmed	
SAP and DMS integration approach confirmed	  <u>Vishey</u> K & Pader
Shipping-line API / fallback approach confirmed	 <u>Vishey</u> K & Pader
Daily, monthly and on-demand Scrap / FA reporting requirements approved	  <u>Vishey</u> K & Pader
24x7 automation availability, support coverage and maintenance approach confirmed	
Security, access-control and audit-trail requirements approved	
Historical documents and expected outcomes available for UAT	
Project risks, assumptions, constraints and open questions reviewed	
Final business, IT and Supervity implementation sign-off completed	



Business Requirement Document (BRD)

for

AGFZE Scrap Trading AI Command Centre

Prepared by: Supervity (Digitamize Innovations Pvt. Ltd.)

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1. Introduction

1.1 Purpose

This Business Requirements Document defines the business scope, stakeholders, current-state challenges, target process, functional requirements, controls, integration expectations, UAT approach and delivery baseline for the AGFZE Scrap & FA Trade Automation Platform.

The document is intended to provide a common reference for AGFZE business teams, IT and integration teams, Supervity implementation teams, approvers and UAT participants. It also establishes the baseline from which detailed functional and technical design can proceed.

1.2 Business Context

AGFZE currently manages document-heavy purchase, sales, logistics and related Scrap / FA trade activities across global transactions. The operating model depends substantially on Outlook emails, Excel trackers, manual validation of invoices and trade documents, and follow-up across suppliers, brokers, customers, logistics users, finance teams and downstream systems.

The target requirement is a web-based AI Command Centre that converts unstructured emails and attachments into structured, validated and trackable business transactions. The solution will classify incoming emails, extract information from invoices, contracts, Bills of Lading and shipping documents, validate data against business rules and source records, create or update transaction entries in the portal, and initiate approved tracker, SAP and DMS actions. The same controlled framework will support the FA business stream, with FA-specific documents, fields and rules finalized during discovery.

1.3 Intended Users

- AGFZE Head of Department (HOD) and designated business approvers.
- Scrap and FA business users across Purchase, Sales, Logistics / Operations and Finance teams.
- AGFZE IT, SAP, DMS, Microsoft 365 and integration-support teams.
- Supervity implementation, AI, integration, quality-assurance and support teams.
- Audit, information-security and UAT stakeholders, where applicable.

1.4 Document Objectives

- Define the agreed business boundary for Scrap and FA purchase, sales, logistics, reporting and document workflow automation.
- Describe the target operating process and the role of AI-assisted extraction, validation, matching, draft preparation, automated entry creation and workflow orchestration.
- Document the business rules, exceptions, data fields, controls and approval gates that the solution must enforce.
- Provide the foundation for solution design, development, testing, UAT, deployment and operational handover.

2. Scope Definition

2.1 In Scope

Scope Area	Included Requirement
Email Intake and Classification	Receipt of supplier, broker, customer, logistics and internal emails through approved Outlook group/shared mailboxes or folders; classification of emails and attachments by transaction type and document category.
Purchase Document Processing	Extraction and validation of supplier invoices, purchase contracts, batch details, container details, quantities, rates, amounts and mandatory support documents.
Sales Document Processing	Generation or validation of sales invoices, contracts and related documents based on OBL, approved draft B/L where applicable, shipping receipt, customer details and transaction references.
FA Transaction Processing	Apply the common intake, extraction, matching, validation, approval, portal-entry and downstream workflow framework to the FA business stream. FA-specific document types, fields, tolerances and approval rules will be finalized during discovery.
Shipment and Cargo Tracking	Tracking of containers, Bills of Lading, shipping status, port details and shipment milestones, including portal / tracker updates and follow-up status.
Web Portal / AI Command Centre	Primary operational interface for transaction status, document review, validation, approvals, exception handling, automated entries, work queues and audit history.
Excel / SharePoint Synchronization	Controlled read / update of Loading Sheets, Shipping Status reports and other operational trackers while they remain part of the transition-state process.
Approvals and Exception Handling	Routing of approvals to designated users and routing of mismatches, missing documents, low-confidence results and validation failures to an exception queue.
Downstream Integration	Triggering of controlled SAP and DMS actions through confirmed APIs or approved integration mechanisms after required validation and approval.
Automated Reporting	Automated daily, monthly and on-demand reports for HOD and designated Scrap / FA users, using portal transaction, approval, shipment, exception and KPI data.
Dashboard and Audit Trail	Visibility of transaction, document, shipment, approval, exception and report status, with a complete action and decision history.

2.2 Out of Scope / Boundary Conditions for the Current Version

- Changes to SAP core configuration, SAP master-data structures or accounting rules.
- Fully autonomous payment execution or irreversible financial commitment without designated AGFZE approval.
- Guaranteed shipping-line API coverage for every carrier. AGFZE will support obtaining available API / portal access where possible; carriers without usable APIs will use an approved fallback method.
- Immediate retirement of all existing Excel / SharePoint trackers. The web portal will be the primary operational view, while controlled synchronization continues during transition until AGFZE approves tracker retirement.
- The specific OCR / AI model is a technical design decision. Exact SAP objects / API names and DMS endpoint contracts will be finalized during detailed integration design without changing the signed business scope.
- Any process, document type, integration or business unit not explicitly validated during requirements sign-off.

2.3 Phasing and Roadmap

- Phase 1: confirm business requirements, Scrap / FA mandatory documents, tolerances, approver matrix, system interfaces, reporting requirements and the W1-W12 calendar baseline.
- Phase 2: configure the AI platform, email and document intelligence, Scrap / FA purchase and sales workflows, web portal, automated entry creation, shipment tracking, exception queues, dashboards and reports.
- Phase 3: complete SAP, DMS, tracker, shipping and notification integrations; execute SIT and UAT.
- Phase 4: production go-live, hypercare, knowledge transfer and handover.

3. Stakeholders and Governance

3.1 Identified Stakeholders

Stakeholder	Primary Responsibility
Head of Department / Business Approver	Approves key documents and final transaction actions; owns business governance and sign-off; reviews automated daily, monthly and on-demand management reports.
Purchase Team	Handles supplier-side document receipt, supplier invoice validation, purchase contract checks and Loading Sheet / portal updates.
Sales Team	Handles customer-side transactions, sales invoice / document generation, customer document validation and sales-record updates.
FA Business Team	Reviews FA transactions and documents, confirms FA-specific business rules and exceptions, and validates FA report requirements.
Logistics / Operations Team	Tracks containers, shipments, Bills of Lading, port status and milestones; resolves logistics exceptions and validates shipment-report information.

Stakeholder	Primary Responsibility
Finance Team	Reviews invoice values, payment readiness and finance-related exceptions before SAP or DMS commitment; supports validation of finance-related report values.
IT / SAP / DMS / Microsoft 365 Support	Maintains connectivity, authentication, APIs, approved group/shared mailbox access, integration monitoring and downstream error handling.
Supervity Implementation Team	Designs, configures and deploys AI agents, extraction logic, portal workflows, business rules, exception queues, dashboards, reports and integration workflows.
Audit / Information Security	Reviews control, access, audit-trail and compliance requirements, where applicable.

3.2 Governance Principles

- The AI and automation layer may classify, extract, match, validate, recommend, prepare drafts, create portal entries, generate reports and trigger approved workflows.
- Final approval for irreversible SAP, DMS or payment-impacting actions remains with the designated AGFZE stakeholder for both Scrap and FA processes.
- Every exception must have an accountable owner, reason, age, priority and next action.
- Every manual correction or override must retain the user, timestamp, reason and source-document reference.
- Automated reports must be traceable to the same governed transaction and audit data used by the portal.

3.3 Draft RACI Matrix

R = Responsible, A = Accountable, C = Consulted, I = Informed. The matrix applies across Scrap and FA workflows where the role is relevant and must be confirmed by AGFZE.

Process Step	Business Approver	Purchase	Sales	Logistics	Finance	IT / SAP / DMS	Supervity
Email intake and request registration	I	C	C	C	I	C	R/A
Document classification and extraction	I	C	C	C	I	C	R/A
Purchase validation and Loading Sheet / portal update	A	R	I	C	C	C	R
Sales document processing	A	I	R	C	C	C	R
Shipment and cargo tracking	I	C	C	R/A	I	C	R
Business-rule validation	A	C	C	C	C	I	R

Process Step	Business Approver	Purchase	Sales	Logistics	Finance	IT / SAP / DMS	Supervity
Exception resolution	A	R	R	R	R	C	R
Business approval	R/A	C	C	C	C	I	C
SAP / DMS / workflow integration	I	I	I	I	C	R/A	R
HOD reporting	A	C	C	C	C	I	R
SIT and integration verification	I	C	C	C	C	R	R/A
UAT and business sign-off	A	R	R	R	R	C	C
Production deployment and hypercare	I	C	C	C	C	R	R/A

3.4 Role-Based Access Control

Role	Permitted Activities
Business Approver	Review transaction summary and evidence; approve, reject, return or request correction; view the approval trail and HOD reports.
Purchase User	Review supplier-side records, correct purchase fields, resolve document or Loading Sheet exceptions and submit for approval.
Sales User	Review customer-side records, prepare or validate sales documents and resolve sales-document exceptions.
FA User	Review FA records, confirm FA-specific fields / documents, resolve exceptions and submit for approval.
Logistics User	Review container, B/L, vessel, port, ETA / ETD and milestone information; resolve tracking exceptions.
Finance User	Review invoice amount, commercial readiness and finance exceptions; validate readiness for downstream action.
IT / Integration Support	Manage connection failures, retries, credentials, integration monitoring and approved manual fallback.
Supervity Support / Administrator	Configure AI policies, extraction rules, routing, queues, portal views, reports and dashboards; support errors without bypassing business approval.
Observer / Auditor	Read-only access to approved dashboards, reports, transaction evidence and audit history.

4. As-Is Process Overview

4.1 Current Operating Process

1. Trade documents and shipment information are received through Outlook from suppliers, brokers, customers, shipping stakeholders and internal users.
2. Business users manually identify the transaction type and document category, then review email content and attachments.
3. Invoice, contract, batch, container, B/L, quantity, rate, amount, port and counterparty details are manually checked against existing trackers and source records.
4. Purchase, Sales, Logistics and Finance teams coordinate through email and manual follow-up to resolve missing documents, mismatches and approvals.
5. Loading Sheets, Shipping Status reports and related Excel trackers are updated manually or through existing workflows.
6. Power Automate, SAP and DMS actions are initiated after the relevant business checks and approvals, depending on available interfaces.
7. Shipment status and document follow-up are monitored through available shipping inputs, manual confirmation and periodic checks.
8. Transaction evidence is distributed across emails, attachments, trackers, approvals and downstream system records.

4.2 Key Challenges

Area	Current-State Challenge
Workflow	Email-based and fragmented across multiple teams and systems.
Validation	Manual, repetitive and dependent on user interpretation.
Tracking	Limited real-time visibility across transaction, document and shipment status.
Documents	Multiple attachments and versions require manual classification and consolidation.
Matching	Transaction references may be inconsistent or missing across invoices, contracts, batches, containers and B/L records.
Exceptions	Missing documents, low-confidence extraction and integration failures require manual follow-up.
Audit	Evidence and decision history are distributed and difficult to reconstruct.
Integration	Target architecture should prefer direct APIs / controlled integrations; Power Automate should not be a primary To-Be dependency and may be retained only for a specifically approved use case such as container tracking if required.
Reporting	Daily / monthly management reporting is manually assembled and not consistently generated from a single governed transaction view.

5. To-Be Process Flow

5.1 Target Operating Model

The target platform will operate as a controlled transaction workflow through a web-based AI Command Centre. Each inbound email or document will be registered, classified, extracted, matched, validated and routed through the required portal, approval and integration gates. The platform will not silently perform irreversible business actions; final approval will remain with the designated AGFZE user.

AGFZE Scrap & FA Trade Automation Platform - Target Process



Human approval remains mandatory before irreversible downstream actions. The web portal is the primary operational view; controlled tracker synchronization continues where required.

Figure 1: High-Level To-Be Process Flow

5.2 High-Level End-to-End Process

9. A supplier, broker, customer, logistics stakeholder or internal user sends an email with trade documents or shipment information to an approved AGFZE group/shared mailbox or folder.
10. The automation layer captures the email, identifies the sender, classifies the request and separates attachments into invoices, contracts, Bills of Lading, shipping documents, trackers and other supporting documents.
11. Document-intelligence agents extract invoice number, contract number, batch number, container number, B/L number, quantity, rate, amount, port and customer / supplier details. FA-specific fields are extracted according to the approved FA checklist.
12. The matching engine links extracted data with open transaction records and available Excel / SharePoint source trackers using identifiers and supporting fields.
13. The validation layer checks mandatory documents, tolerance rules, field consistency, duplicates / repeated information, confidence and approval requirements.
14. When validations pass, the platform creates or updates the governed transaction entry in the web portal and prepares any approved draft documents.
15. When validations fail, the item is moved to an exception queue with owner, issue reason, ageing and next action.
16. Required business approval is captured before committed SAP, DMS or other irreversible downstream updates.

17. Shipment and container status is updated through available shipping-line APIs, approved portal / assisted checks or manual confirmation.
18. Daily, monthly and on-demand reports plus dashboards and audit logs retain the complete record of source email, documents, extracted fields, rules, approvals, exceptions, downstream results and report outputs.

5.3 Process States

State	Business Meaning
Received	Email and attachments captured; classification not yet completed.
Classified	Email category and document types identified.
Extracted	Supported fields extracted and normalized with source evidence.
Matched	Request linked to an existing transaction or a new transaction shell.
Validation Pending	Business rules and document checks are running.
Needs Review / Exception	Missing, ambiguous, low-confidence, repeated or mismatched information requires action.
Portal Record / Draft Ready	Validated transaction entry and any allowed draft document have been created / prepared in the portal.
Approval Pending	Transaction is ready for the designated approver.
Approved	Required human approval has been captured for the next action.
Integration Pending	Tracker, SAP, DMS or approved integration action is queued or in progress.
Committed / Closed	Downstream result captured and no further transaction action remains; reporting / audit data retained.

6. Functional Requirements

6.1 Email Intake and Document Classification

Req. ID	Requirement / Activity	Primary Owner	Expected Output
FR-EI-01	Email is received in Outlook from a supplier, broker, customer, shipper or internal user through an approved group/shared mailbox or folder.	System / Business User	Email source, timestamp, subject, body and attachments captured.
FR-EI-02	System reads email body and attachment metadata.	AI Email Agent	Email categorized as purchase, sales, FA, logistics, approval, follow-up, informational or exception.
FR-EI-03	Attachments are classified into supported document types.	AI Classification Agent	Invoice, contract, B/L, shipping document, tracker, approval support, FA document or unknown document identified.
FR-EI-04	Mandatory document checklist is applied according to transaction type.	Rule Engine	Missing-document condition flagged before downstream processing.
FR-EI-05	A unique Request ID and transaction shell are created.	System	Request registered for audit and tracking.
FR-EI-06	The AI Agent is granted approved access to the existing AGFZE group/shared mailbox where this is the selected intake route. A new dedicated mailbox is not mandatory.	IT / Microsoft 365 / System	Authorized mailbox access established without requiring unnecessary mailbox duplication.

6.2 Purchase-Side Document Processing

Req. ID	Requirement / Activity	Primary Owner	Expected Output
FR-PU-01	Receive supplier invoice, purchase contract, loading details and supporting documents.	Purchase / System	Input pack registered through the approved group/shared mailbox, folder or portal intake route.
FR-PU-02	Extract supplier, invoice, contract, batch, quantity, rate, amount, currency, container and port details.	Extraction Agent	Structured purchase transaction data available.
FR-PU-03	Match extracted data with Loading Sheet or existing transaction records.	Matching Agent	Batch, container or contract reference linked to an open purchase record.

Req. ID	Requirement / Activity	Primary Owner	Expected Output
FR-PU-04	Validate mandatory fields, quantity tolerance, invoice value, repeated / duplicate information and document completeness.	Validation Agent	Valid record proceeds; mismatch routed to exception.
FR-PU-05	Create / update the purchase transaction entry in the web portal and synchronize the Loading Sheet where required after successful validation.	System / Workflow	Portal record plus controlled tracker status, validation status and next action recorded.
FR-PU-06	Route approval when transaction value, document status or exception rules require review.	Purchase / Approver	Approval decision and trail captured before downstream action.

6.3 FA-Side Transaction Processing

Req. ID	Requirement / Activity	Primary Owner	Expected Output
FR-FA-01	Receive FA transaction emails and supporting documents through the approved group/shared mailbox, folder or portal intake route.	FA / System	FA input pack registered against a Request ID.
FR-FA-02	Classify and extract the approved FA document and commercial fields.	AI Classification / Extraction Agent	Structured FA transaction data available with source evidence.
FR-FA-03	Match FA documents to the relevant transaction, contract, batch, counterparty or shipment reference.	Matching Agent	FA record linked to the correct transaction.
FR-FA-04	Validate FA mandatory documents, configured tolerances, duplicate / repeated information and approval conditions.	Validation Agent	Validated FA item proceeds; mismatch is routed to exception.
FR-FA-05	Create or update the FA transaction entry in the web portal and prepare downstream entries / documents where applicable.	System / FA User	Governed FA record and draft / downstream-ready data created.
FR-FA-06	Obtain designated business approval before irreversible SAP / DMS action.	FA Approver / HOD	Approval status and audit trail captured.

FA-specific document types, fields, tolerances and approval rules are business inputs to be finalized with AGFZE during discovery. The common platform controls above are in scope now.

6.4 Sales-Side Document Processing

Req. ID	Requirement / Activity	Primary Owner	Expected Output
FR-SA-01	Receive OBL, shipping confirmation, approved draft B/L or other sales-trigger document.	Sales / Logistics / System	Sales trigger identified and linked to a transaction.
FR-SA-02	Extract customer, sales contract, B/L, container, quantity, rate, shipment and port details.	Extraction Agent	Structured sales and shipment data available.
FR-SA-03	Match extracted details with purchase / cargo records and the sales record / tracker.	Matching Agent	Sales record linked to purchase batch, container or contract reference.
FR-SA-04	Validate sales-invoice readiness and mandatory supporting documents.	Validation Agent	Missing required document, mismatched B/L or unmatched container routed to exception.
FR-SA-05	Generate or prepare the sales invoice, contract or shipment-related document using approved templates. Where AGFZE permits, a draft may be prepared based on an approved draft B/L; final committed action requires the required final shipping validation.	System / Sales	Draft or completed document produced and linked to source evidence.
FR-SA-06	Obtain final approval before downstream posting or DMS upload.	Sales Approver / HOD	Approved status captured.

6.5 Shipment and Cargo Tracking

Req. ID	Requirement / Activity	Primary Owner	Expected Output
FR-LG-01	Capture container, B/L, shipping line, vessel, port and shipment milestone information from documents, portal records or trackers.	Logistics / System	Shipment record created or updated.
FR-LG-02	Attempt scheduled tracking updates using available shipping-line APIs or other approved mechanisms. AGFZE will support obtaining available carrier access where possible.	Shipment Tracking Agent	Latest status, ETA / ETD and milestones captured where available.
FR-LG-03	Update the portal shipment view and synchronize the Shipping Status report where required with validated container and milestone information.	System / Workflow	Business users receive current shipment visibility.
FR-LG-04	Route delayed, missing or inconsistent shipment status for follow-up.	Logistics / Exception Agent	Owner, ageing and next action assigned.
FR-LG-05	Link shipment status to the associated purchase, sales and FA transaction records where relevant.	System	Unified transaction, container and document status.

6.6 Approval, Portal Entry, Tracker, SAP and DMS Workflow

Req. ID	Requirement / Activity	Primary Owner	Expected Output
FR-IN-01	Route the validated transaction for required business approval.	System / Approver	Approver, timestamp, decision and remarks captured.
FR-IN-02	Create or update the validated transaction entry in the web portal; synchronize approved trackers where they remain required.	System / Workflow	Governed portal record and controlled tracker reference captured.
FR-IN-03	Trigger an SAP update through the confirmed API or approved integration mechanism when eligible.	System / SAP Support	SAP reference or failure reason logged.
FR-IN-04	Upload approved documents and supporting attachments to DMS.	System / DMS Support	DMS document ID or link captured.
FR-IN-05	Route integration failures to a technical or business exception queue.	System / IT Support	Retry, approved workaround or escalation action assigned.

6.7 Dashboard and Workbench Requirements

Req. ID	Requirement
FR-DB-01	Provide status visibility for pending, processed, failed and exception transactions.
FR-DB-02	Show exception ageing, owner, priority, reason and next action.
FR-DB-03	Show document completeness, validation status, approval status and shipment status.
FR-DB-04	Show automation percentage, extraction accuracy, processing turnaround and integration failure counts.
FR-DB-05	Provide review queues for low-confidence extraction, ambiguous matching, approvals and technical failures.
FR-DB-06	Allow users to open the source email, attachment, portal transaction and relevant tracker / system reference from the transaction record, subject to access rights.
FR-DB-07	Use the web portal / AI Command Centre as the primary operational interface for Scrap and FA users, with role-based access to transaction and workbench views.

6.8 Automated Reporting Requirements

Req. ID	Requirement
FR-RP-01	Generate a daily operational report for the HOD and designated Scrap / FA stakeholders using governed portal data.
FR-RP-02	Generate a monthly management summary for Scrap and FA with transaction status, approvals, exceptions, ageing, shipment visibility and agreed KPIs.
FR-RP-03	Allow authorized users to generate on-demand / ad-hoc reports by date range, business stream (Scrap / FA), transaction status and other approved filters.
FR-RP-04	Ensure report figures are traceable to the underlying transaction, source documents and audit trail.
FR-RP-05	Make reports available through the web portal and support configured distribution to approved recipients / channels. Exact report templates and distribution rules will be confirmed with AGFZE.

7. Business Rules, Exceptions and Data Requirements

7.1 Key Business Rules

Rule ID	Business Rule
BR-01	Every transaction must have a unique Request ID created from email / portal and attachment intake.
BR-02	Invoice number, contract number, batch number, container number, B/L number or another approved business reference must be available before transaction matching can be completed.
BR-03	Container number must match the transaction record before purchase, sales or shipment-status updates are committed.
BR-04	Mandatory documents such as invoice, contract, B/L, shipping support or approved FA documents must be present according to transaction type.
BR-05	Quantity variation is permitted only within a configured tolerance. The current draft uses plus or minus five percent as an indicative value, subject to AGFZE confirmation.
BR-06	Invoice value, rate, quantity and amount must match the approved source record or fall within configured tolerance.
BR-07	Sales-document preparation may be triggered by an OBL or, where explicitly approved by AGFZE rules, a draft B/L. Final committed posting / document issuance must wait for the required final shipping document validation.
BR-08	Missing documents, mismatched identifiers, out-of-tolerance values and duplicate / repeated references must move to an exception or review queue.
BR-09	The web portal is the primary operational transaction view. Excel / SharePoint trackers are synchronized only after successful extraction, matching and validation and only where they remain required during transition.
BR-10	For both Scrap and FA, SAP or DMS updates must not be committed until required business approval is captured.
BR-11	AI extraction outputs must retain confidence score, source-document reference and manual override history.
BR-12	Workflow actions, approvals, exceptions, report generation and downstream updates must be audit logged.
BR-13	Repeated emails / documents must not create duplicate downstream postings. New or changed information must be linked to the existing transaction and processed according to version / update rules.

7.2 Exception Handling Matrix

Exception	System / Process Action	Owner
Missing mandatory document	Pause processing, mark request incomplete and notify the sender / owner with the missing-document list.	Purchase / Sales / FA / Logistics
Mismatched container number	Block portal / tracker update and route to transaction owner for confirmation or correction.	Logistics / Operations
Invoice amount outside tolerance	Hold invoice processing and route to Finance / transaction owner for review.	Finance / Purchase / Sales / FA
Quantity variation outside tolerance	Require approval or corrected document before processing continues.	Purchase / Sales / FA / Approver
Unmatched contract / batch / B/L reference	Move to matching-review queue and request reference clarification.	Purchase / Sales / FA / Logistics
Low OCR / AI confidence	Route to human review with highlighted field and source-document reference.	Business User / Supervity Support
Duplicate / repeated document or information	Link to the existing transaction, compare new versus existing information, block duplicate posting and retain the attempt / version in the audit trail.	System / Process Owner
Shipment status unavailable or delayed	Create follow-up task, assign owner and update ageing in the shipment dashboard.	Logistics / Operations
Approval not received	Keep in pending-approval status and escalate according to the approved ageing rule.	Approver / HOD
SAP / DMS / integration failure	Log technical failure, route to support queue and retry or use an approved fallback.	IT / System Support

7.3 Required Data Fields

Data Domain	Required Fields
Purchase	Request ID; supplier name; supplier invoice number; purchase contract number; batch number; quantity; rate; amount; currency; Loading Sheet reference; mandatory-document status.
Sales	Request ID; customer name; sales invoice number; sales contract number; batch / transaction reference; quantity; rate; amount; currency; OBL / draft or final B/L / shipping-trigger reference; mandatory-document status.
FA	Request ID; counterparty; transaction / contract / batch reference; quantity; rate; amount; currency; document type; approval status; additional FA-specific fields to be finalized during discovery.
Shipment / Logistics	Request ID; container number; B/L number; shipping line; vessel; port of loading; port of discharge; ETA; ETD; milestone; shipment status.
Control / Workflow	Email received date; sender / mailbox; document type; AI confidence score; validation status; exception reason; approval status; workflow run ID; portal / tracker-update timestamp; SAP reference; DMS link; report-generation reference.

7.4 Automation / Workflow Requirements

Automation Area	Requirement
Approved Mailbox / Folder	AGFZE trade emails may be processed from an approved existing group/shared mailbox or designated folder. A new dedicated mailbox is not mandatory.
Email Classification	Classify emails into purchase, sales, FA, logistics, approval, exception, follow-up and informational categories.
Document Classification	Classify attachments as invoice, contract, Bill of Lading, shipping document, FA document, tracker, approval evidence or unknown.
Data Extraction	Extract invoice, contract, batch, container, B/L, supplier / customer, quantity, rate, amount, currency, port and shipment details plus approved FA fields.
Rule Engine	Make completeness, tolerance, matching, duplicate / repeated information, approval and eligibility rules configurable.
Transaction Matching	Match documents to records using batch, container, contract, invoice, B/L or shipment references.
Web Portal Entry	Create / update governed transaction records, work queues, approvals and exceptions in the AI Command Centre.
Excel / SharePoint Synchronization	Read / update validated records through controlled Microsoft Graph or approved integration mechanisms where trackers remain in use.

Automation Area	Requirement
SAP / DMS	Trigger the confirmed downstream mechanism after validation and approval. Direct APIs / controlled integration are preferred.
Exception Queue	Store owner, reason, ageing, priority, source-document link and next action.
Reporting	Generate daily, monthly and on-demand HOD reports from governed transaction data.
Audit Trail	Log every email, attachment, extraction, validation, approval, override, report generation, workflow trigger and system response.

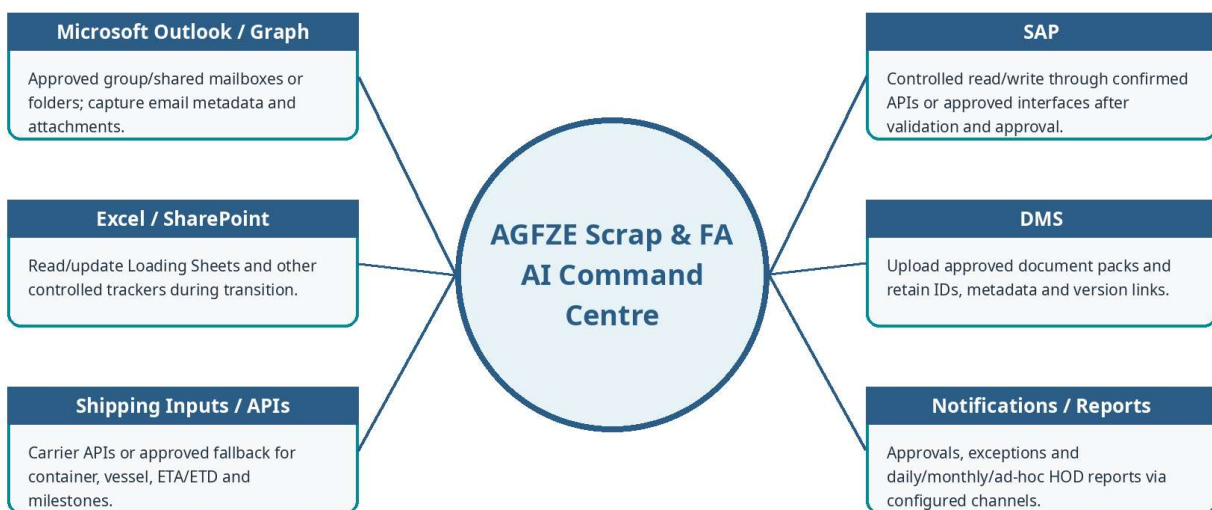
8. Integration Requirements

System	Business Requirement	Required Confirmation / Input
Microsoft Outlook / Graph	Read approved group/shared mailboxes or folders, capture message metadata and attachments, and preserve the source message reference.	Mailbox / app access, folder rules, security approval and AGFZE decision on group/shared mailbox routing.
Web Portal / AI Command Centre	Create and maintain governed transaction records, queues, approvals, exceptions, report outputs and audit references.	Role matrix, portal fields, view requirements and authentication approach.
Excel / SharePoint	Read and update Loading Sheets, Shipping Status reports and other controlled trackers during transition.	Workbook locations, table / sheet definitions, field mappings, access and retirement / transition decision.
SAP	Read or update approved transaction objects through confirmed BAPI, RFC, OData, API or another approved controlled interface.	Exact objects, credentials, environments, API contracts and posting gates.
Document Management System	Upload approved document packs, capture document IDs / links and preserve metadata and version information.	Endpoint, metadata schema, folder structure and retention rules.
Shipping Inputs / APIs	Retrieve vessel, container, ETA / ETD, port and milestone status where available. AGFZE will support obtaining carrier access / credentials where possible.	Carrier coverage, credentials, rate limits and fallback method.
Notifications / Reporting	Send approval, exception, escalation and completion notifications; distribute configured daily / monthly / ad-hoc reports.	Recipient rules, templates, channels, report schedule and escalation timings.

8.1 Integration Design Principles

- Use authenticated and controlled interfaces approved by AGFZE IT.
- Prefer direct APIs / controlled integrations. Power Automate is not a primary To-Be dependency and may be used only for a specifically approved use case if required.
- Prevent duplicate updates through transaction and action identifiers and repeated-information checks.
- Capture request, response, timestamp, status and error details for every downstream action.
- Support retries and an approved manual or assisted fallback where a target system is unavailable.
- Keep SAP core configuration unchanged unless a separately approved change is raised.

AGFZE Scrap & FA AI Command Centre - Integration Landscape



Design principle: direct APIs / controlled integrations are preferred. Power Automate is not a primary dependency and may be used only for a specifically approved use case if required.

Figure 2: Updated Integration Landscape

9. Non-Functional Requirements

NFR ID / Area	Requirement
NFR-01 Performance	Target processing should complete within a few minutes after document receipt when source systems and integrations are available. Exact SLA and percentile targets are to be confirmed.
NFR-02 Availability	The automated platform should support 24x7 processing and monitoring, subject to planned maintenance and the availability of dependent source / target systems. Business support and escalation coverage may follow separately agreed support hours.
NFR-03 Scalability	The solution should scale for high document volumes and concurrent transactions. Expected daily / monthly volumes and peak-load assumptions are to be provided.
NFR-04 Security	Data must be encrypted in transit and at rest; access must be role based and logged.
NFR-05 Audit and Compliance	Emails, attachments, extraction results, validations, approvals, overrides, reports and downstream responses must be retained in an auditable sequence.
NFR-06 Reliability and Error Handling	Failed integrations must be retryable, visible and routed to a support queue without duplicating downstream actions.
NFR-07 Data Quality and Explainability	Extracted fields must retain confidence and source evidence; low-confidence or ambiguous outcomes must be reviewable.
NFR-08 Maintainability	Business rules, tolerance values, document checklists, routing, report schedules and notification templates should be configurable without uncontrolled code changes.
NFR-09 Compatibility	The solution will support the agreed business document types and browser / Microsoft 365 / SAP / DMS environments. The final maximum per-file / attachment size will be confirmed during technical design using representative AGFZE files and target-system constraints; oversized items must be flagged for controlled handling rather than failing silently.
NFR-10 Retention and Recovery	Data-retention, archival, backup, recovery-point and recovery-time requirements are to be confirmed by AGFZE.

9.1 Control Requirements

Control	Requirement
Maker-Checker	Automated extraction and validation must remain separate from final business approval.
Role-Based Access	Users may access only the transactions, dashboards, reports, trackers and documents relevant to their role.
Duplicate / Repeated Information Prevention	Duplicate invoices, contracts, containers, B/Ls, emails, repeated information and workflow submissions must be detected and prevented from causing duplicate downstream actions. Updated information must be linked to the existing transaction with version history.
Tolerance Control	Quantity and amount tolerance values must be configurable and visible in validation and audit output.
Document Completeness	Mandatory-document checklist must be applied before portal finalization, tracker synchronization, SAP or DMS updates.
SAP Clean Core	SAP integration should use controlled interfaces or approved assisted automation without unauthorized core changes.
Audit Logging	Source email, document, extraction, validation, approval, override, report and downstream results must be retained.
KPI Tracking	Automation percentage, extraction accuracy, turnaround, shipment visibility, exception ageing and reporting delivery must be measured.

10. User Acceptance Testing Strategy

10.1 UAT Approach

UAT will validate that the platform processes representative Scrap and FA purchase, sales, logistics, reporting and approval scenarios in accordance with approved requirements, business rules, approval gates and integration behavior. Historical AGFZE documents should be used to prove extraction, matching, validation, portal updates, controlled tracker synchronization, exception routing, reporting and downstream handoff.

10.2 High-Level UAT Scenarios

Scenario ID	Scenario
UAT-01	Correct classification of purchase, sales, FA, logistics, approval and exception emails.
UAT-02	Correct classification and extraction of invoices, contracts, B/L, FA and shipping documents.
UAT-03	Successful purchase matching against Loading Sheet / transaction references.
UAT-04	Successful sales matching based on OBL, B/L, container, contract or batch reference.
UAT-05	Mandatory-document, quantity, rate, amount, repeated / duplicate information and tolerance validation.
UAT-06	Low-confidence and ambiguous cases routed to the correct review queue.
UAT-07	Approval, rejection, return, correction and resubmission workflow.
UAT-08	Validated update to the web portal and controlled Excel / SharePoint synchronization without duplicate or conflicting write.
UAT-09	Successful and failed SAP / DMS / integration with visible response and retry behavior.
UAT-10	Shipment-status update, delayed-status exception and operational dashboard visibility.
UAT-11	Audit reconstruction from source email through final downstream result and report.
UAT-12	Role-based access prevents unauthorized actions.
UAT-13	Automated daily, monthly and on-demand report creation for Scrap and FA teams / HOD using governed portal data.
UAT-14	Approved draft B/L scenario: draft sales document can be prepared where allowed, while final committed action remains blocked until required final shipping validation.

10.3 Test Data Requirements

- Historical purchase invoices, purchase contracts, loading records and supporting documents.
- Historical sales invoices, contracts, OBL / final B/L and representative approved draft B/L scenarios, and related customer / bank documents where applicable.
- Representative FA documents and expected FA transaction outputs once the FA checklist is confirmed.
- Representative shipment records with containers, ports, ETA / ETD and milestone updates.
- Positive, negative, duplicate / repeated, missing-document, low-confidence, unmatched and out-of-tolerance cases.

- Expected outcomes and business-approved reference values for each test case.
- Expected daily / monthly report examples and validation totals for Scrap and FA.

10.4 Proposed Entry Criteria

- Requirements, rules, FA checklist and approval matrix approved for UAT scope.
- UAT environment, user access, integrations and test group/shared mailboxes available.
- Historical test documents and expected results prepared.
- System Integration Testing completed with no unresolved critical defect.

10.5 Proposed Exit Criteria

- All critical scenarios executed with evidence.
- No open Severity 1 defect and no unresolved defect that prevents the agreed business process.
- Business owners approve known limitations, workarounds and production-readiness actions.
- Formal AGFZE UAT sign-off captured before production deployment.

11. Implementation Plan and Timeline

11.1 Planning Basis

The implementation plan uses the 12-week Gantt baseline provided for the Scrap Trading AI Command Center and has been updated to include FA workflow coverage and enhanced reporting. It establishes the sequence for discovery, platform build, workflow automation, integrations, testing, go-live and hypercare. Calendar dates must be mapped to W1-W12 during project kickoff.

- Requirements and solution-architecture sign-off are targeted for W3.
- SAP and DMS interface readiness is targeted for W4.
- AI platform readiness is targeted for W5.
- Purchase, sales and FA workflow completion is targeted for W6.
- SAP and DMS integration completion is targeted for W8.
- Dashboard and reporting readiness is targeted for W9.
- SIT sign-off is targeted for W9 and UAT sign-off for W10.
- Production go-live is targeted for W11, followed by hypercare closure in W12.

11.2 Milestone Baseline

Milestone	Target Week	Owner
Project Kickoff	W1	Supervity + AGFZE
Requirements Sign-off	W3	AGFZE
Solution Architecture Sign-off	W3	Supervity / AGFZE
SAP and DMS Readiness	W4	SAP Team / DMS Team
AI Platform Ready	W5	Supervity
Purchase Workflow Complete	W6	Supervity

Milestone	Target Week	Owner
Sales Workflow Complete	W6	Supervity
FA Workflow Complete	W6	Supervity / AGFZE
SAP Integration Complete	W8	SAP Team + Supervity
DMS Integration Complete	W8	DMS Team + Supervity
Dashboard & Reporting Ready	W9	Supervity / AGFZE
SIT Sign-off	W9	QA / Integration Teams
UAT Sign-off	W10	AGFZE
Production Go-Live	W11	Supervity + AGFZE
Hypercare Closure	W12	Supervity + AGFZE

11.3 Twelve-Week Delivery Plan

12-Week Delivery Plan - Updated Baseline

#	Workstream	Owner	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	Status
□	Project Kickoff	Supervity / AGFZE													Milestone
1	Discovery & Planning - Business	AGFZE / Supervity													In Progress
2	Discovery & Planning - Integration	SAP / DMS / Supervity													Scheduled
3	AI Platform Build	Supervity AI Team													Scheduled
4	Purchase Workflow Automation	Supervity / SAP													Scheduled
5	Sales Workflow Automation	Supervity / SAP													Scheduled
6	FA Workflow Automation	Supervity / AGFZE / SAP													Scheduled
7	Document Intelligence	Supervity AI Team													Scheduled
8	SAP Integration	SAP / Supervity													Scheduled
9	DMS Integration	DMS / Supervity													Scheduled
10	AI Workflow Orchestration	Supervity AI Team													Scheduled
11	Dashboard & Reporting	Supervity / AGFZE													Scheduled
12	System Integration Testing	QA / SAP / DMS													Scheduled
13	User Acceptance Testing	AGFZE / Supervity													Scheduled
14	Deployment & Hypercare	Supervity / AGFZE													Scheduled
□	Production Go-Live	Supervity + AGFZE													Milestone

FA workflow is added within the existing 12-week baseline; exact sequencing and calendar dates remain subject to kickoff confirmation.

Figure 3: Updated Twelve-Week Delivery Plan

11.4 Workstream Scope and Ownership

#	Workstream	Description	Owner	Status
1	Discovery & Planning - Business	Finalize requirements, validate As-Is and To-Be workflows, FA scope, reporting requirements, success criteria and business sign-off.	AGFZE / Supervity	In Progress
2	Discovery & Planning - Integration	Confirm SAP, DMS, Microsoft 365, portal, tracker and shipping integration scope, access and security approvals.	SAP / DMS / Supervity	Scheduled
3	AI Platform Build	Configure the command centre, AI runtime, workflow routing, roles, queues and notifications.	Supervity AI Team	Scheduled
4	Purchase Workflow Automation	Configure supplier invoice, contract, Loading Sheet / portal, validation, approval and exception flows.	Supervity / SAP	Scheduled
5	Sales Workflow Automation	Configure OBL / draft-final B/L, sales document, transaction matching, approval and exception flows.	Supervity / SAP	Scheduled
6	FA Workflow Automation	Configure approved FA document intake, extraction, matching, validation, portal entry, approvals and exceptions.	Supervity / AGFZE / SAP	Scheduled
7	Document Intelligence	Configure document classification, OCR / extraction, field evidence, confidence and document packaging.	Supervity AI Team	Scheduled
8	SAP Integration	Implement confirmed read / write actions and response handling.	SAP / Supervity	Scheduled
9	DMS Integration	Implement upload, metadata, link capture, versioning and failure handling.	DMS / Supervity	Scheduled
10	AI Workflow Orchestration	Implement end-to-end routing, approval gates, business rules, alerts and audit trail.	Supervity AI Team	Scheduled
11	Dashboard & Reporting	Implement operational portal views, KPIs, shipment visibility, approvals, exception ageing and daily / monthly / ad-hoc Scrap & FA reports.	Supervity / AGFZE	Scheduled
12	Testing & Go-Live	Execute SIT, UAT, defect resolution, readiness review, deployment and hypercare.	AGFZE / Supervity / Integration Teams	Scheduled

11.5 Delivery Dependencies

- Approved requirements, mandatory-document checklists, FA business rules, tolerance rules and templates.
- Access to test and production group/shared mailboxes or folders, representative documents and existing Excel / SharePoint trackers while they remain part of transition-state integration.
- Confirmed portal field model and role / access requirements.
- Confirmed SAP and DMS integration pattern, interfaces and credentials.
- Available shipping-line API / portal access supported by AGFZE where possible.
- Approved report templates, schedules, recipients and expected totals / KPIs.
- AGFZE business availability for clarification, review, SIT support, UAT and sign-off.
- Infrastructure, security and network approvals completed in line with the delivery plan.

12. Assumptions and Dependencies

ID	Assumption / Dependency
A-01	An approved existing Outlook group/shared mailbox or designated folder may be used for each intake route. A new dedicated mailbox is not mandatory, subject to Microsoft 365 / security approval.
A-02	AGFZE will provide representative historical documents, tracker samples and expected results for validation and UAT.
A-03	Approved document templates, mandatory-document checklists and naming conventions will be provided.
A-04	AGFZE will confirm quantity and amount tolerances, confidence thresholds, approval conditions and escalation timing.
A-05	Required access, credentials, APIs and integration support will be available according to the agreed timeline.
A-06	The web portal will be the primary operational view. Existing trackers remain operational during transition where required and are synchronized through controlled workflows until AGFZE approves retirement.
A-07	Business approval remains mandatory before irreversible downstream actions.
A-08	Automated processing is expected to operate 24x7 subject to planned maintenance and dependent system availability; business support coverage will be agreed separately.
A-09	AGFZE will support obtaining shipping-line API / portal access where available; fallback methods will be agreed for carriers without usable APIs.
A-10	AGFZE will confirm FA-specific document types, field mappings, tolerances and approval rules during discovery.
A-11	Expected transaction volume, peak load, retention period, report recipients and final file-size limits will be confirmed before production sizing.

13. Constraints

ID	Constraint
C-01	Shipping-line status coverage depends on the API, portal, website or manual mechanism available for each carrier.
C-02	Document quality, scan quality and inconsistent layouts may affect extraction confidence and require human review.
C-03	The To-Be solution should not depend on Power Automate where direct APIs / controlled integrations are available. Power Automate may be considered only for a specifically approved use case, such as container tracking, if required.
C-04	Excel / SharePoint trackers may still be edited by users during transition while automation is processing, requiring conflict detection, revalidation and controlled synchronization.

ID	Constraint
C-05	The twelve-week schedule is dependent on timely client inputs, API readiness, security approval and stakeholder availability.
C-06	Any new document category or business workflow discovered after scope sign-off may require change control.
C-07	Final FA-specific data and rule completeness depends on AGFZE confirmation during discovery.

14. Risks and Mitigation

Risk ID	Risk	Impact	Mitigation
R-01	Delayed SAP / DMS access or API readiness	High	Confirm owners and readiness dates during W1-W3; use a tested mock / approved fallback where needed.
R-02	Incomplete or changing business rules, including FA rules	High	Freeze mandatory documents, tolerances, FA rules and approval matrix before build completion; manage changes through a controlled log.
R-03	Poor document quality or inconsistent formats	Medium	Use confidence thresholds, source evidence and human-review queues; expand training samples during UAT.
R-04	Ambiguous transaction references	High	Use layered matching and require manual link / confirmation for weak matches.
R-05	Duplicate / repeated or conflicting portal / tracker / system updates	High	Use unique request and action identifiers, repeated-information checks, current-record re-read and before / after audit logging.
R-06	User adoption and exception backlog	Medium	Provide role-based training, queue ownership, ageing KPIs and hypercare support.
R-07	Shipping / SAP / DMS API unavailability requires a fallback integration pattern	High	Prefer direct APIs; confirm approved fallback by W3-W4. Use Power Automate only for a specifically approved use case if needed.
R-08	Compressed testing and go-live timeline	High	Protect SIT / UAT entry criteria, prioritize critical scenarios and maintain a formal readiness decision.
R-09	Report totals differ from source transactions due to transition-state tracker conflicts	Medium	Generate reports from governed portal records, reconcile against source systems during UAT and retain report-to-transaction traceability.

15. Glossary

Term	Definition
AGFZE	AGFZE business entity / project stakeholder.
AI	Artificial Intelligence.
B/L	Bill of Lading.
DMS	Document Management System.
ETA / ETD	Estimated Time of Arrival / Estimated Time of Departure.
FA	AGFZE FA business stream / team referenced in the approved scope.
HOD	Head of Department.
OBL	Original Bill of Lading.
OCR	Optical Character Recognition.
RACI	Responsible, Accountable, Consulted and Informed.
SAP	Enterprise Resource Planning system used for downstream business actions.
SIT	System Integration Testing.
UAT	User Acceptance Testing.